MIL-I-51319A
15 August 1983
SUPERSEDING
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3 December 1968

MILITARY SPECIFICATION

INSECTICIDE, PYRETHRIN, SPACE SPRAY, SYNERGIZED PYRETHRINS

This specification is approved for use by all
Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers one solvent base space spray containing synergized pyrethrins (hereinafter referred to as space spray).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. Unless otherwise specified, the following specifications, standards, and handbooks of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

O-P-921	-	Pyrethrum Extract
NN-P-71	-	Pallets, Material Handling, Wood, Stringer Construction, 2-Way and 4-Way (Partial)
TT-E-485	-	Enamel, Semigloss, Rust-Inhibiting
PPP-B-585	-	Boxes, Wood, Wirebound
PPP-B-601	-	Boxes, Wood, Cleated-Plywood
PPP-B-621	-	Boxes, Wood, Nailed and Lock-Corner
PPP-C-96	-	Cans, Metal, 28 Gage and Lighter
PPP-F-320	-	Fiberboard; Corrugated and Solid, Sheet Stock (Container Grade), and Cut Shapes

: Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed: to: Commander, Armament Research and Development Center, US Army Armament, Munitions and Chemical Command, ATTN: DRSMC-TSC-S(A), Aberdeen Proving Ground, MD 21010 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

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STANDARDS

FEDERAL

Fed. Std. No. 595 - Colors

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129 - Marking for Shipment and Storage
MIL-STD-147 - Palletized Unit Loads
MIL-STD-1168 - Ammunition Lot Numbering

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 162 - Regulations for the Enforcement of the Federal Insecticide, Fungicide, and Rodenticide Act
49 CFR 171 to 179 - Department of Transportation Hazardous Materials Regulations

(The Code of Federal Regulations is available from the Superintendent of Documents, US Government Printing Office, Washington, DC 20402. Orders for the above publications should cite "40 CFR 162" or "49 CFR 171 to 179.")

(Copies of specifications, standards, handbooks, drawings, and publications required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following documents form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

ASTM STANDARDS

D484 - Hydrocarbon Drycleaning Solvents
D1193 - Reagent Water
D3951 - Commercial Packaging

(Application for copies should be addressed to ASTM, 1916 Race Street, Philadelphia, PA 19103.)

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ASSOCIATION OF OFFICIAL ANALYTICAL CHEMISTS (AOAC)

Official Methods of Analysis, 10th Edition, 1965.

(Application for copies should be addressed to the Association of Official Analytical Chemists, Incorporated, P.O. Box 540, Benjamin Franklin Station, Washington, DC 20044.)

UNIFORM FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the Uniform Classification Committee, Room 1106, 222 South Riverside Plaza, Chicago, IL 60606.)

NATIONAL MOTOR FREIGHT CLASSIFICATION RULES

(Application for copies should be addressed to the American Trucking Associations, Inc., Traffic Department, 1616 P Street, NW, Washington, DC 20036.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Composition. The space spray shall be a thoroughly blended solution of pyrethum extract, synergist, and deodorized kerosene, in the proportions specified in table I (see 4.2.3.4).

TABLE I. Composition

Material	Percent by weight	Reference paragraph
Pyrethrum extract, minimum	2.0	3.1.1
Piperonyl butoxide, technical, (synergist), minimum	1.6	3.1.2
Deodorized kerosene, maximum	balance	3.1.3

3.1.2 Synergist characteristics. The piperonyl butoxide, technical used in the manufacture of space spray shall conform to the physical and chemical characteristics of table II when tested as specified therein.

3.1.3 Deodorized kerosene. The deodorized kerosene used to manufacture space spray shall conform to type II of ASTM D484.

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TABLE II. Physical and chemical characteristics

Characteristic	Requirement	Test paragraph
Specific gravity (20°C)	1.04 to 1.07	*
Refractive index (20°C)	1.480 to 1.500	**
Assay content, percent by weight	80	4.2.4.1

*Determined by any suitable method accurate to 0.001
**Determined by Abbe refractometer or other suitable instrument

3.2 Pyrethrins content of space spray. The manufactured space spray shall contain no less than 0.4 percent by weight of pyrethrins when tested as specified in 4.2.4.2.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Quality conformance inspection.

4.2.1 Lotting. A lot shall consist of the space spray produced by one manufacturer, at one plant, from the same materials, and under essentially the same manufacturing conditions provided the operation is continuous. In the event the process is a batch operation, each batch shall constitute a lot (see 6.3). Each lot shall be identified and controlled in accordance with MIL-STD-1168.

4.2.2 Sampling.

4.2.2.1 For examination of packaging. Sampling shall be conducted in accordance with MIL-STD-105.

4.2.2.2 For pyrethrins content (see 6.5). Sampling shall be conducted in accordance with table III. A representative specimen of approximately 250 grams (g) shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

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TABLE III. Sampling for test

Number of containers in batch or lot	Number of sample containers
36 to 150	3
151 to 1,200	5
1,201 to 7,000	8
7,001 to 20,000	10
Over 20,000	20

4.2.2.3 For synergist (see 6.5). The lot or batch of synergist used to manufacture space spray shall be sampled prior to the manufacture of the space spray in accordance with table III. A representative specimen of approximately 10 g shall be removed from each sample container and placed in a suitable clean, dry container labeled to identify the lot and container from which it was taken.

4.2.3 Inspection procedure.

4.2.3.1 For examination of packaging. The sample unit shall be one filled unit or shipping container, as applicable, ready for shipment. Sample unit and shipping containers shall be examined for the following defects using an AQL of 2.5 percent defective:

(a) Contents per container not as specified
(b) Container not as specified
(c) Container closure not as specified
(d) Container damaged or leaking
(e) Fiberboard pads missing or not as specified (when required)
(f) Unitization not as specified
(g) Marking incorrect, missing, or illegible

4.2.3.2 For pyrethrins content. Each sample specimen taken in 4.2.2.2 shall be tested as specified in 4.2.4. Failure of any test by any specimen shall be cause for rejection of the lot represented.

4.2.3.3 For synergist. Each sample specimen taken in 4.2.2.3 shall be tested as specified in 4.2.4.1. If the specimen fails either analysis, the lot represented shall not be used to manufacture space spray.

4.2.3.4 For composition. Weight records of the amount and lot number of each ingredient used to manufacture each lot of space spray shall be kept for the information of an inspector.

4.2.4 Tests (see 6.5). Water in accordance with ASTM D1193 and reagent grade chemicals shall be used throughout the tests. Where applicable, blank determinations shall be run and corrections applied where significant. Tests shall be conducted as follows:

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4.2.4.1 Piperonyl butoxide, technical. Determine the percent by weight of alpha[2-(2-butoxyethoxy)ethoxy]-4,5-(methylenedioxy)-2-propyltoluene in piperonyl butoxide, technical in accordance with AOAC methods 4.256 through 4.259.

4.2.4.2 Pyrethrins content of space spray. Determine the percent by weight pyrethrins in the manufactured space spray in accordance with the applicable test method specified in O-P-921.

5. PACKAGING

5.1 Preservation. Preservation shall be level A or industrial, as specified (see 6.2).

5.1.1 Level A. One US gallon of space spray shall be unit packed in a nominal 1-gallon capacity oblong can formed from tin-plate and fitted with a metal screw cap conforming to type V, class 4 of PPP-C-96. Each can shall be provided with exterior nonmetallic coating in accordance with plan B of PPP-C-96, and shall have side seam striped with enamel conforming to TT-E-485. The metal screw cap liner shall be provided with a vinyl facing and each closure shall be provided with an inner seal as specified in PPP-C-96. The filled cans shall be tightly closed and neither the can nor its closure shall leak when tested as specified in PPP-C-96.

5.1.2 Industrial. One US gallon of insecticide shall be unit packed in accordance with ASTM D3951.

5.2 Packing. Packing shall be level A, B, or industrial as specified (see 6.2).

5.2.1 Level A. Six cans of space spray, unit packed as specified in 5.1.1, shall be packed in a nailed wood, cleated plywood, or wirebound wood box as specified (see 6.2). The nailed wood box shall conform to class 2, style 4, grade A, for a type 2 average load of PPP-B-621. The cleated plywood box shall conform to type overseas, style A, B, I, or J, grade A, for a type 2 average load of PPP-B-601. The wirebound wood box shall conform to class 3, style optional, for a type 2 average load of PPP-B-585; and the wood parts shall be preserved as specified for the grade A box of PPP-B-621. Half slotted fiberboard separators of the same height as the cans (+1/8 or -0 inch) shall be inserted in each box to separate the cans. The size of the cells formed by the separators shall be such that free movement of the cans within each cell is prevented. All inside faces of each box shall be lined with full box-size fiberboard pads in sufficient quantity to form a tight pack. Fiberboard separators and pads shall be formed of material conforming to grade W5c of PPP-F-320. Each box shall be closed and strapped using galvanized strapping as specified in the appendix to the applicable box specification.

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5.2.2 Level B. Packing shall be as specified in 5.1.1 except that the box shall conform to style RSC, grade V3c of PPP-B-636. The box shall be closed by taping in accordance with the requirements of PPP-B-636 for closure of class weather resistant boxes, style RSC, to be shipped as exterior containers. Each box shall be reinforced by pressure-sensitive adhesive filament banding as specified in PPP-B-636.

5.2.3 Industrial. The space spray shall be packed in accordance with ASTM D3951.

5.3 Marking. Level A and B containers of space spray shall be marked in accordance with MIL-STD-129. Industrial containers shall be marked in accordance with ASTM D3951.

5.3.1 Labeling. Each container shall be labeled in compliance with 40 CFR 162 and 49 CFR 171 to 179. Labels shall be approved and registered in accordance with 40 CFR 162. The following labeling data, as applicable, shall be lithographed on the side of each unit container using noncorrosive, durable marking material of yellow color No. 23793 or No. 33793 of Fed. Std. No. 595. Alternatively, paper labels may be used when specified (see 6.2), applied as specified in MIL-STD-129:

FSC 6840-
INSECTICIDE, PYRETHRUM, SPACE SPRAY, SYNERGIZED PYRETHRINS
1 US Gallon

For use against flies, mosquitoes,
and similar small flying insets.

Ingredient statement (see 5.3.1.1)

CAUTION

Keep out of the reach of children.
Avoid contamination of feed and foodstuffs.
Avoid prolonged or repeated inhalation of mist.
Avoid contact with eyes, skin, and clothing.
Harmful if swallowed.
Wash thoroughly after handling.

FIRST AID: In case of contact, immediately flush eyes with plenty of water for at least 15 minutes. Call a physician. Flush skin with water. (Wash clothing before reuse.) If swallowed and victim is conscious, have victim drink water or milk. Do not induce vomiting. Call a physician.

Keep away from heat and open flame.
In case of fire, extinguish with dry chemical, foam, or carbon dioxide. Water may be ineffective on fire. Cool exposed container with water.

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DIRECTIONS FOR USE

To control flies and mosquitoes, close room and spray into the air. Adjust the nozzle of the sprayer to give the finest mist possible. Spray at a rate of 3/4 to 1 fluid ounce per 1000 cubic feet or until a moderate mist can be seen in all parts of the room. It is not necessary to spray the air heavily. Keep the room closed for 10 minutes.

Manufacturer's name and address

Lot number Date of pack

5.3.1.1 Ingredient statement. The ingredient statement shall be as follows:

Active ingredients	Percent by weight
Pyrethrins	0.4
Technical piperonyl butoxide (equivalent to 1.28 percent (butyl-carbityl) (6-propylpiperonyl) ether and 0.32 percent related compounds)	1.6
Deodorized kerosene	96.4

5.4 Unitization. Level A and B packs of space spray shall be palletized in accordance with load type I of MIL-STD-147 using the softwood pallet conforming to type IV, size 2 of NN-P-71. Industrial packs shall be unitized to conform to Uniform Freight Classification Rules, National Motor Freight Classification Rules or the rules of any other applicable carrier, and to assure acceptance by the carrier and safe delivery to destination.

6. NOTES

6.1 Intended use. Space spray is intended for use against flies, mosquitoes, and similar small flying insects in an inclosed area.

6.2 Ordering data. Acquisition documents should specify the following:

(a) Title, number, and date of this specification
(b) Level of preservation and packing required (see 5.1 and 5.2)
(c) Kind of box required for level A packing (see 5.2.1)
(d) If paper labels may be used (see 5.4.1)

6.3 Batch. A batch is defined as that quantity of material which has been manufactured by some unit chemical process or subjected to some physical mixing operation intended to make the final product substantially uniform.

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6.4 Significant places. For the purpose of determining conformance with this specification, an observed or calculated value should be rounded off "to the nearest unit" in the last right-hand place of figures used in expressing the limiting value, in accordance with the rounding-off method of ASTM E29.

6.5 Sampling and testing precautions. This specification covers inspection of chemical material (insecticide, pyrethrin, space spray, synergized pyrethrins) which is potentially hazardous to personnel. All applicable safety rules, regulations and procedures shall be followed in the handling and processing of these materials.

6.6 Synergistic compounds. A synergistic compound is an ingredient which enhances or increases the effect of the active agent.

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Army - GL, MD
DLA - GS

User activities:

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Preparing activity:

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions -- Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-I-51319A	INSECTICIDE, PYRETHRIN, SPACE SPRAY, SYNERGIZED PYRETHRINS

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR
b. ADDRESS *(Street, City, State, ZIP Code)*	☐ USER
	☐ MANUFACTURER
	☐ OTHER *(Specify)*:

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation.

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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